AWS Solution Architect

(UNIT-2:Compute & Storage)

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Core Infrastructure Services of AWS

- **Compute**: Provides the processing power to run applications, with services like Amazon EC2 for virtual servers, AWS Lambda for serverless code execution, AWS Fargate for container compute, and AWS Elastic Beanstalk for deploying web applications.
- **Storage**: Offers reliable and scalable options including Amazon S3 for object storage, Amazon EBS for block storage volumes, Amazon EFS for shared file storage, and Amazon S3 Glacier for archive storage.
- **Networking & Content Delivery**: Services for network environments and efficient content delivery, such as Amazon VPC for isolated virtual networks, Amazon Route 53 for DNS, Amazon CloudFront for a global CDN, Elastic Load Balancing for distributing traffic, and AWS Direct Connect for dedicated network connections to AWS.
- **Databases**: A variety of managed database services including Amazon RDS for relational databases, Amazon DynamoDB for NoSQL, Amazon Aurora for high-performance relational databases, and Amazon Redshift for data warehousing.

Amazon EC2 (Elastic Compute Cloud)

- Amazon EC2 is like renting a powerful, virtual computer in the cloud that you can use to run any application.
- Instead of buying and managing your own physical servers, you can rent these "instances" from Amazon, customize their hardware like processor and memory, and pay only for what you use.
- **Key concepts**
- Virtual computers: EC2 provides virtual machines, called "instances," that act as your own private servers.
- On-demand capacity: You can get compute capacity whenever you need it, and the amount can be adjusted "elastically" to match your workload.
- Scalability: You can quickly add or remove instances to handle more or less traffic, a feature known as auto-scaling.

Amazon EC2 (Elastic Compute Cloud)

- Amazon Elastic Compute Cloud (Amazon EC2) is a web service which is provided by the AWS cloud which is secure, resizable, and scalable.
- You can deploy your applications in EC2 servers without worrying about the underlying infrastructure.
- You configure the EC2-Instance in a very secure manner by using the VPC (Virtual Private Cloud), Subnets, and Security groups.
- You can scale the configuration of the EC2 instance you have configured based on the demand of the application by attaching the autoscaling group to the EC2 instance.
- You can scale up and scale down the instance based on the incoming traffic of the application.

Amazon EC2 (Elastic Compute Cloud)

- Provide secure and resizable compute capacity in cloud by making it easy for the developers.

Features

- Scaling
- Integration with other services like S3 (Simple Storage Service), etc,.
- Pay as per you use
- Instances can be launched in one or more AZ's and regions
- Supports different operating systems
- Works with amazon VPC to provide secure networks

Use Cases of Amazon EC2 (Elastic Compute Cloud)

The following are the use cases of Amazon EC2:

- **Deploying Application:** In the AWS EC2 instance, you can deploy your application like .jar, .war, or .ear application without maintaining the underlying infrastructure.
- **Scaling Application:** Once you deployed your web application in the EC2 instance know you can scale your application based upon the demand you are having by scaling the AWS EC2-Instance.
- **Deploying The ML Models:** You can train and deploy your ML models in the EC2-instance because it offers up to 400 Gbps, and storage services purpose-built to optimize the price performance for ML projects.
- **Hybrid Cloud Environment:** You can deploy your web application in EC2-Instance and you can connect to the database which is deployed in the on-premises servers.
- **Cost-Effective:** Amazon EC2-instance is cost-effective so you can deploy your gaming application in the Amazon EC2-Instances.

AWS EC2 Instance Types

The AWS EC2 Instance types are as follows:

- General Purpose Instances
- Compute Optimized Instances
- Memory-Optimized Instances
- Storage Optimized Instances
- Accelerated Computing Instances

AWS EC2 Instance Types

1. **General Purpose Instances**

- It provides the balanced resources for a wide range of workloads.
- It is suitable for web servers, development environments, and small databases.
- Examples: T3, M5, etc,.

T3 stands for Burstable Performance General Purpose Instance. T3 instances are a type of Amazon EC2 instance designed to provide a balance of compute, memory, and networking resources for general-purpose applications that experience moderate CPU usage with occasional spikes.

2. Compute Optimized Instances

- It provides high-performance processors for compute-intensive applications.
- It will be Ideal for high-performance web servers, scientific modeling, and batch processing.
- Examples: C5, C6g instances.

AWS EC2 Instance Types

3. Memory-Optimized Instances

- High memory-to-CPU ratios for large data sets.
- Perfect for in-memory databases, real-time big data analytics, and high-performance computing (HPC).
- Examples: R5, X1e instances.

4. Storage Optimized Instances

- It provides optimized resource of instance for high, sequential read and write access to large data sets.
- Best for data warehousing, Hadoop, and distributed file systems.
- Examples: I3, D2 instances.

AWS EC2 Instance Types

5. Accelerated Computing Instances

- It facilitates with providing hardware accelerators or co-processors for graphics processing and parallel computations.
- It is ideal for machine learning, gaming, and 3D rendering.
- Examples: P3, G4 instances.

Features of AWS EC2 (Elastic Compute Cloud)

The following are the features of AWS EC2:

1. AWS EC2 Functionality

- EC2 provides its users with a true virtual computing platform, where they can use various operations and even launch another EC2 instance from this virtually created environment. This will increase the security of the virtual devices.
- Not only creating but also EC2 allows us to customize our environment as per our requirements, at any point of time during the life span of the virtual machine.
- Amazon EC2 itself comes with a set of default AMI(Amazon Machine Image) options supporting various operating systems along with some pre-configured resources like RAM, ROM, storage, etc.
- Besides these AMI options, we can also create an AMI curated with a combination of default and user-defined configurations. And for future purposes, we can store this user-defined AMI, so that next time, the user won't have to re-configure a new AMI(Amazon Machine Image) from scratch. Rather than this whole process, the user can simply use the older reference while creating a new EC2 machine.

Features of AWS EC2 (Elastic Compute Cloud)

2. AWS EC2 Operating Systems

- Amazon EC2 includes a wide range of operating systems to choose from while selecting your AMI.
- Not only are these selected options, but users are also even given the privilege to upload their own operating systems and opt for that while selecting AMI during launching an EC2 instance.
- Currently, AWS has the following most preferred set of operating systems available on the EC2 console.

 Amazon Linux

 Windows Server

 Ubuntu Server

 SUSE Linux

 Red Hat Linux

Features of AWS EC2 (Elastic Compute Cloud)

3. AWS EC2 Software

- Amazon is single-handedly ruling the cloud computing market, because of the variety of options available on EC2 for its users.
- It allows its users to choose from various software present to run on their EC2 machines.
- This whole service is allocated to AWS Marketplace on the AWS platform. Numerous software like SAP, LAMP, Drupal, etc are available on AWS to use.
- SAP: (Systems, Applications, and Products in Data Processing)
- LAMP: Linux, Apache, MySQL, and PHP

Features of AWS EC2 (Elastic Compute Cloud)

4. AWS EC2 Scalability and Reliability

- EC2 provides us the facility to scale up or scale down as per the needs.
- All dynamic scenarios can be easily tackled by EC2 with the help of this feature. And because of the flexibility of volumes and snapshots, it is highly reliable for its users.
- Due to the scalable nature of the machine, many organizations like Flipkart, and Amazon rely on these days whenever humongous traffic occurs on their portals.

EC2 Pricing Models

Amazon EC2 offers four pricing options for instances: On-Demand Instances, Reserved Instances, Spot Instances, and Dedicated Hosts.

- **On-Demand Instances:** The On-Demand instance is like a pay-as-you-go model where you have to pay only for the time you are going to use if the instance is stopped then the billing for that instance will be stopped when it was in the running state then you are going to be charged. The billing will be done based on the time EC2-Instance is running.
- **Reserved Instances (Saving Plans)**: Reversed Instance is like you are going to give the commitment to the AWS by buying the instance for one year or more than one year by the requirement to your organization. Because you are giving one year of Commitment to the AWS they will discount the price on that instance.
- **Spot Instances:** You have to bid the instances and who will win the bid they are going to get the instance for use but you can't save the data which is used in this type of instance. Here you can get upto 90% of discount compared to On-Demand instances
- **Dedicated Host Reservation:** It provides you with a discount of up to 70% compared to On-Demand pricing. Here, you pay for each active dedicated host you use, regardless of the number or size of instances running on that host,

EC2 Pricing Models

Pricing Model	Description	Billing Method	Use Case	Key Features
On-Demand Instances	Pay for compute capacity per hour or per second (for Linux instances).	Per-hour or per-second	Best for short-term or variable workloads.	No upfront payments. Flexible, pay only for what you use. Easy scaling.
Spot Instances	Leverage unused EC2 capacity at a discount of up to 90%.	Per-hour or per-second	Suitable for cost-sensitive applications with flexible start/end times or large-scale workloads.	Pricing fluctuates based on supply and demand. Ideal for workloads tolerant of interruptions.
Reserved Instances	Discount of up to 75% compared to On-Demand Instances when you commit to a 1 or 3-year term.	Per-hour	Recommended for predictable workloads.	Capacity reservation in a specific Availability Zone. Long-term commitment results in cost savings.
Dedicated Hosts	A physical EC2 server reserved exclusively for your use.	Per-hour	Best for customers with compliance requirements or those using existing software licenses.	Helps with compliance requirements. Utilize existing software licenses (e.g., Windows Server, SQL Server).

EC2 Pricing Models

Per-Second Billing Scheme For Amazon EC2

- Many customers utilize Amazon EC2 to accomplish significant tasks in a very short time, sometimes just minutes or even seconds. In 2017, AWS introduced per-second billing for Linux instance usage across On-Demand, Reserved, and Spot Instances. The minimum billing duration is one minute (60 seconds), but after that initial minute, charges are applied on a per-second basis. However, if you start and then stop an instance within 30 seconds, you will still be billed for the full 60 seconds.
- This per-second billing option is available for all instances launched in:

- On-Demand Instances
- Reserved Instances
- Spot instances
- All regions
- All Availability Zones
- Amazon Linux
- Windows
- Ubuntu

Key Components Of Amazon EC2 Costs

When estimating the costs associated with Amazon EC2, it's essential to consider several key components:

- **Server Time Charges:** Billing for EC2 resources begins the moment your instances are launched and continues until they are terminated. Additionally, charges apply from the time an Elastic IP address is allocated until it is released.
- **Instance Type Selection:**EC2 instance types consist of various combinations of resources, such as CPU, memory, networking capabilities, and storage options. You can scale instance sizes according to the specific needs of your workload, allowing for customization and optimization.
- **Number of Instances:** The pricing model for AWS EC2 is influenced by the number of instances you are running simultaneously. Provisioning multiple instances can impact overall costs.
- **Elastic Load Balancing:** To manage traffic effectively, AWS provides Elastic Load Balancing, which distributes incoming traffic across your EC2 instances. The total hours that Elastic Load Balancing is active will contribute to your monthly expenses.

Key Components Of Amazon EC2 Costs

- **Monitoring Features:** Amazon EC2 integrates with Amazon CloudWatch for monitoring capabilities. While basic monitoring is included by default, opting for detailed monitoring incurs a fixed monthly fee.
- **Elastic IP Addresses:** AWS offers one Elastic IP address per running instance at no additional cost. However, any additional Elastic IPs may lead to extra charges.
- **Software Licensing:** AWS provides flexibility in software licensing, allowing you to either obtain licenses through AWS or bring your own. AWS licenses operate on a “pay-as-you-go” basis, while using your own licenses can lower total cost of ownership (TCO), provided the agreements align with cloud use. AWS License Manager can assist in managing your software licenses effectively.

AWS Elastic IP Addresses

- An AWS Elastic IP address is a static, public IPv4 address designed for dynamic cloud computing, meaning it can be programmatically associated with an instance in your account.
- Unlike the dynamic IP that an Amazon EC2 instance gets by default, which can change if the instance is stopped and restarted, an Elastic IP remains constant and provides a stable public endpoint for your application.
- This allows you to remap the IP to a different instance to handle failover if one instance fails, ensuring consistent availability for your services

Working of AWS Elastic IP

The working of AWS Elastic IP can be explained in the following manner:

1. **Allocate an Elastic IP Address:** It is easy to assign an IP address to your account. Your IP address may be assigned using either the pool of IP addresses offered by Amazon or the private IP address pool you have added to your account.

There are the following options for the Public IPv4 address pool:

- IPv4 Addresses are available on Amazon: If you want an IPv4 address assigned from the pool of IPv4 numbers maintained by Amazon.
- My Collection of Open IPv4 Addresses: If you wish to assign an IPv4 address from a pool of IP addresses you've added to your AWS account. If your network does not contain any IP address pools, this option is disabled.
- The IPv4 Address Pool that belongs to the Customer: If you need to distribute an IPv4 address for use with AWS Outposts from a pool made using your on-premises network. If you don't have AWS Outposts, this option isn't available to you.

Working of AWS Elastic IP

2. **Describe your Elastic IP Address:** In this step, one can view all the metadata of the Elastic IP Address by visiting the View Information Portal. Here the metadata includes the name, descriptions, port numbers assigned, and other necessary details.

3. **Tag an Elastic IP Address:** You can use tags to distinguish between your IP addresses in case you have created multiple IP addresses.

4. **Connect or Disconnect** an Elastic IP address from a Network Interface or Instance

5. **Release an Elastic IP Address**: We advise releasing an Elastic IP address if you no longer require it. The IP that needs to be released cannot currently be connected to any EC2 instance, or Network Load Balancer on AWS.

Features of AWS Elastic IP

1. Static Public IPv4 Address

- An Elastic IP is a **permanent** public IP address that remains constant until you manually release it.
- Unlike dynamically assigned public IPs, an Elastic IP does not change on instance stop/start.

2. Failover & High Availability

- You can **remap** an Elastic IP to another EC2 instance in case of failure.
- This helps ensure **business continuity** by quickly switching traffic to a backup instance.

3. Dynamic Reassignment

- You can **detach and reassign** an Elastic IP from one instance to another **instantly**.
- This allows for **load balancing** and failover solutions without changing DNS settings

Features of AWS Elastic IP

4. One Free Elastic IP Per Running Instance

- AWS provides **one free Elastic IP** per **running EC2 instance**.
- However, **if the Elastic IP is not associated with an instance, AWS charges you for it.**

5. Elastic IP Pooling

- You can **allocate multiple Elastic IPs** and manage them in a pool.
- Useful for large-scale architectures needing multiple static public IPs.

6. IPv4-Only

- Elastic IPs are available **only for IPv4 addresses**, as AWS does not provide Elastic IPv6.
- AWS provides IPv6 addresses differently (e.g., Egress-Only Internet Gateway for IPv6)

AWS Elastic Beanstalk

- AWS Elastic Beanstalk is a managed platform-as-a-service (PaaS) that simplifies the deployment and management of applications in the AWS Cloud. It allows developers to upload their application code, and Elastic Beanstalk automatically handles the underlying infrastructure provisioning, deployment, scaling, and monitoring.
- AWS Elastic Beanstalk uses Amazon EC2 as a foundational service, but it is a higher-level Platform as a Service (PaaS) that manages the EC2 instances for you. When you deploy an application with Elastic Beanstalk, it automatically provisions and configures the necessary AWS resources, including EC2 instances, load balancing, auto-scaling, and monitoring, to run your code
- In essence, Elastic Beanstalk streamlines the process of getting web applications and services up and running on AWS without requiring deep expertise in configuring and managing individual AWS services.

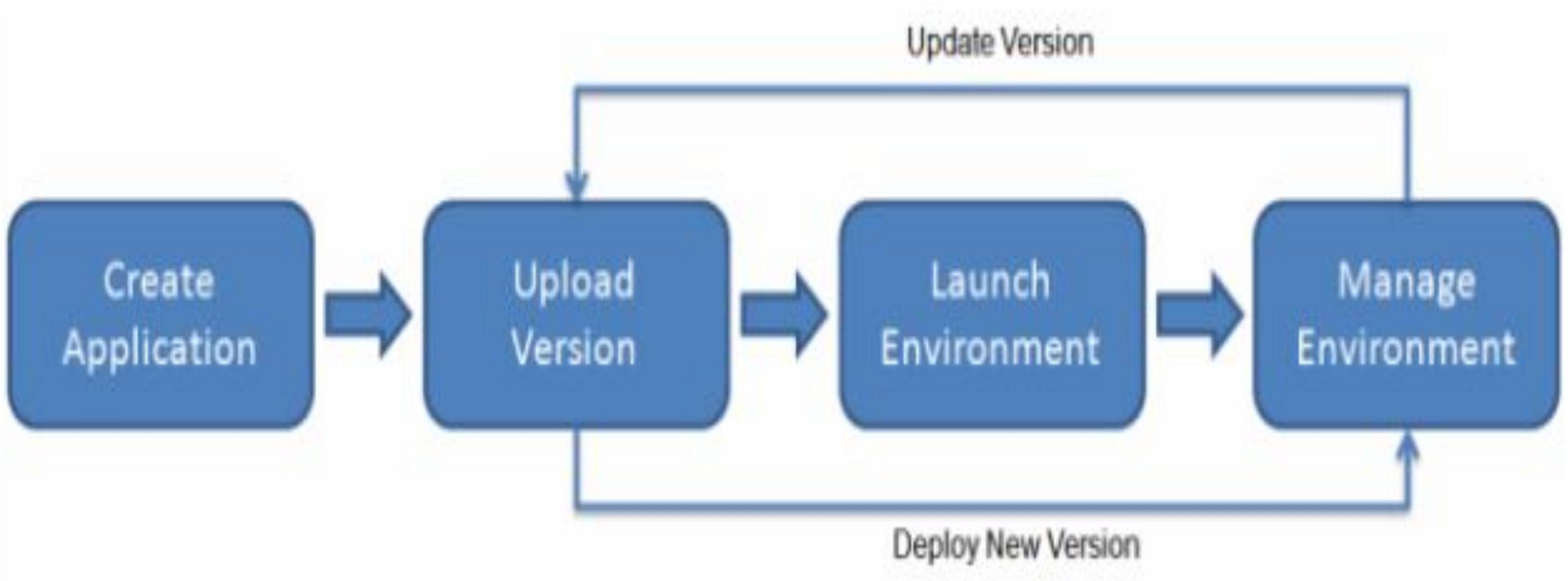
AWS Elastic Beanstalk

- AWS Elastic Beanstalk is an AWS-managed service for web applications.
- Elastic Beanstalk is a pre-configured EC2 server that can directly take up your application code and environment configurations and use it to automatically provision and deploy the required resources within AWS to run the web application.
- Unlike EC2 which is Infrastructure as a service, Elastic Beanstalk is a Platform As A Service (PAAS) as it allows users to directly use a pre-configured server for their application.
- Of course, you can deploy applications without ever having to use elastic beanstalk but that would mean having to choose the appropriate service from the vast array of services offered by AWS, manually provisioning these AWS resources, and stitching them up together to form a complete web application.
- Elastic Beanstalk abstracts the underlying configuration work and allows you as a user to focus on more pressing matters.

How Elastic Beanstalk Works ?

- To use Elastic Beanstalk, you create an application, upload an application version in the form of an application source bundle (for example, a Java .war file) to Elastic Beanstalk, and then provide some information about the application.
- Elastic Beanstalk automatically launches an environment and creates and configures the AWS resources needed to run your code. After your environment is launched, you can then manage your environment and deploy new application versions.
- After you create and deploy your application, information about the application—including metrics, events, and environment status—is available through the Elastic Beanstalk console, APIs, or Command Line Interfaces, including the unified AWS CLI.

How Elastic Beanstalk Works ?



Elastic Beanstalk Features

- Elastic Beanstalk offers preconfigured runtime-like environments and deployment tools which makes it easy to deploy our application
- It supports numerous platforms and programming languages like Golang, Python java, etc.
- Elastic Beanstalk scales your application automatically when the demand increases with the help of auto-scaling rules.
- Elastic Beanstalk can integrate with databases such as Mysql, Oracle, and Microsoft SQL Server.
- Access control via AWS Identity and Access Management and built-in security features like SSL/TLS encryption are provided by Elastic Beanstalk (IAM).

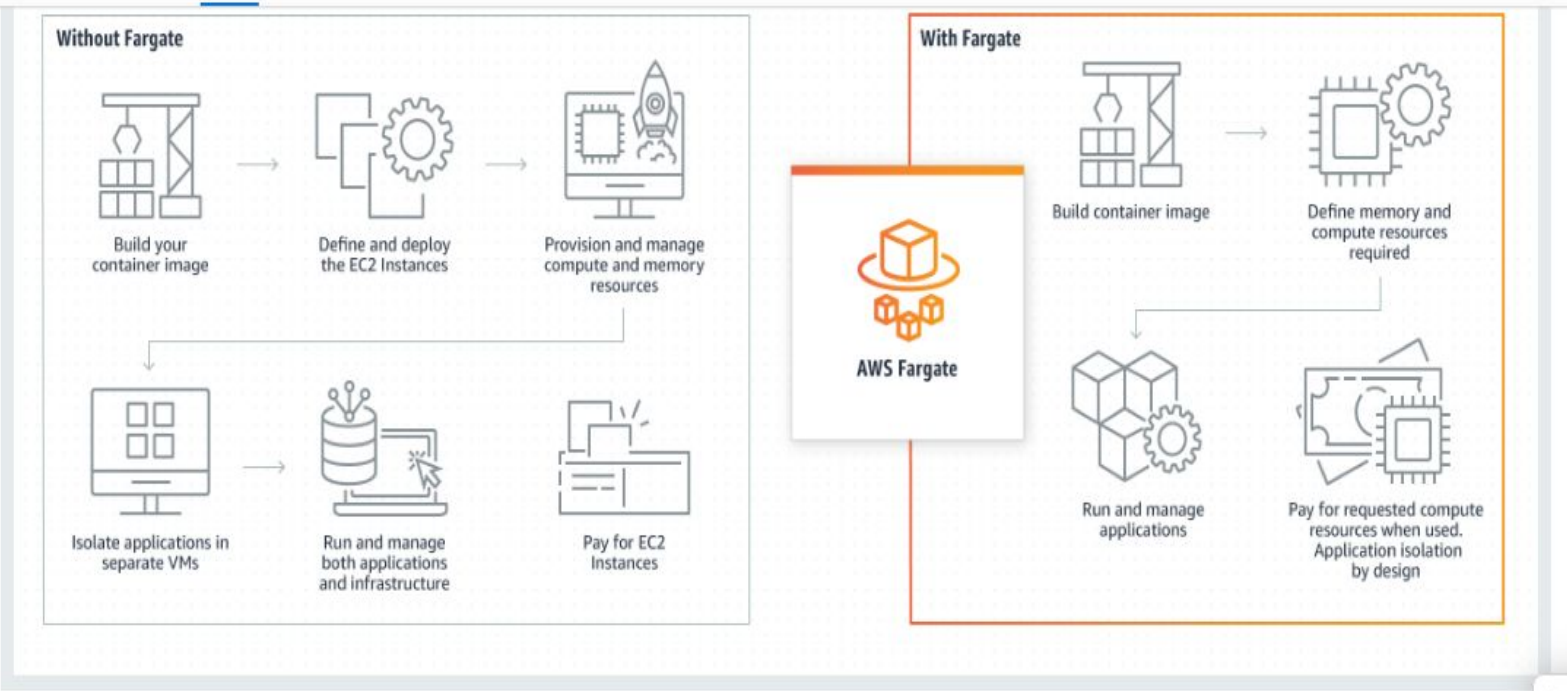
AWS Fargate

- AWS Fargate is a serverless way to run containers, meaning you don't have to manage the underlying servers or clusters yourself. You define your application in containers, specify the resources it needs (like CPU and memory), and AWS handles all the server provisioning, scaling, and management for you.
- No server management: You don't need to provision, configure, or scale virtual machines (\(EC2\)) instances). AWS manages all the underlying infrastructure and clusters.
- Focus on your application: Since AWS handles the servers, you can focus on building and deploying your applications instead of managing infrastructure.
- Automatic scaling: Fargate automatically scales your application to meet demand without you having to manually adjust the cluster size.
- Security through isolation: Each Fargate task runs in its own isolated environment, preventing it from sharing resources like the CPU or kernel with other tasks.

AWS Fargate

- AWS Fargate is a feature in container services in Amazon Web Services that can be used to run your containers without having to manage the server or the underlying architecture.
- It is a serverless computing engine for containers. It is a pay-as-you-go model, i.e. pay for the resources you are using.
- It is one of the most important features of serverless computing in AWS. Whenever you opt for Fargate, the infrastructure will be owned and managed by AWS(Amazon Web Services).
- It saves you time by automating features like patching, updating, and dealing with resources.
- By running the application in the form of containers by using AWS ECS integrating with the ECS service AWS Fargate will take care of everything like server types, decide when to scale your clusters, or optimize cluster packing.

AWS Fargate



Without Fargate
Build your container image
Define and deploy the EC2 Instances
Provision and manage compute and memory resources
Isolate applications in separate VMs
Run and manage both applications and infrastructure
Pay for EC2 Instances
AWS Fargate
With Fargate
Build container image
Define memory and compute resources required
Run and manage applications
Pay for requested compute resources when used. Application isolation by design

AWS Fargate

- AWS offers Fargate as a service which is used to run the deploy or run the application in the form of containers of serverless compute engine. Here you do not need to worry about how to manage the servers it all will be taken care by the AWS itself.
- You can integrate the AWS Fargate with both Elastic Kubernetes services and Elastic container service.
- To use AWS Fargate, package your application into a Docker image, configure the necessary settings like CPU, memory, and networking, and let Fargate handle the rest.
- Fargate runs containers in isolation within virtual machines (VMs) and charges you only for the resources consumed by your containers.

How AWS Fargate Works?

1. Define Your Container

- Use **Docker** to package your application into a container.

2. **Choose Elastic container service(ECS)or Elastic Kubernetes services (EKS)**

- Deploy on **Amazon ECS** (simpler) or **Amazon EKS** (for Kubernetes workloads).

3. Fargate Launch Type

- Select **Fargate** instead of EC2 for running containers.

4. AWS Manages Infrastructure

- AWS **allocates the right CPU, memory, and networking** automatically.

5. Scale Automatically

- Containers **scale up/down** based on demand.

Components Of AWS Fargate

- **Clusters:** A cluster is a logical group of resources (containers) where your applications run. It can contain **multiple services and tasks** running simultaneously.
- **Task Definitions:** These are text files where the user can describe one or more containers that form the applications. They are used to define various specifications or parameters to be used in Fargate containers. They can be called the blueprint of your Fargate application.
- **Tasks:** A task is the initialization or instantiation of the task definition. It is a running instance of a containerized application. Users can create multiple tasks from the same task definition in a cluster. Users can run a single task or multiple tasks at a time.
- **Services:** Services is a tool in Fargate to run the tasks that have been created. A **service** ensures that a specified number of **tasks are always running.** This can be used to run multiple tasks at a time in a cluster. They also replace the tasks with new tasks based on task definition if the existing task fails or stops.

Amazon Elastic Container Service (ECS)

- Amazon Elastic Container Service (ECS) is a fully managed container service that enables users to run Docker-based applications (A Docker container is a lightweight, standalone, and executable software package that includes everything needed to run an application) in containers across a cluster of EC2 instances.
- ECS simplifies automated management of **containerized applications** at scale, allowing you to deploy, manage, and scale containerized applications efficiently.
- The service supports multiple launch types, including the AWS Fargate launch type, which abstracts the underlying EC2 infrastructure, enabling you to run containers without needing to manage or provision the EC2 instances. This allows developers to focus on building and deploying applications while AWS handles the cluster management and resource provisioning. It is best for **auto-scaling & simplified deployments**.
- ECS can also be deployed with EC2 where it runs containers on **self-managed EC2 instances**. It has more control over resources. It is best for **fine-tuned performance & cost control.**

How does ECS work?

- Containerization: You package your application and its dependencies into a "container" (like a portable, self-contained box).
- ECS Manager: ECS is the "manager" that takes these containers and decides which servers to run them on.
- Scalability: If your application gets very busy, ECS can automatically create more copies of your container to handle the load. When it's quiet, it can remove the extra copies to save resources.
- Serverless option (Fargate): With Fargate, you just give ECS the container, and it runs it for you. You don't have to think about managing any virtual servers at all.
- Server option (EC2): You can choose to have ECS run your containers on a group of virtual servers (EC2 instances) that you manage. This gives you more control over the server environment.
- Other AWS services: ECS works seamlessly with other Amazon services, such as automatically directing traffic to your running applications or monitoring their performance.

Amazon Elastic Container Service (ECS)

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- **There are three layers in Amazon ECS:**

- **Capacity** - The infrastructure where your containers run
- **Controller** - Deploy and manage your applications that run on the containers
- **Provisioning** - The tools that you can use to interface with the scheduler to deploy and manage your applications and containers.

- **Amazon ECS capacity**

Amazon ECS capacity is the infrastructure where your containers run. The following is an overview of the capacity options:

- **Amazon EC2 instances in the AWS cloud**

You choose the instance type, the number of instances, and manage the capacity.

- **Serverless (AWS Fargate) in the AWS cloud**

Fargate is a serverless, pay-as-you-go compute engine. With Fargate you don't need to manage servers, handle capacity planning, or isolate container workloads for security.

- **On-premises virtual machines (VM) or servers**

Amazon ECS Anywhere provides support for registering an external instance such as an on-premises server or virtual machine (VM), to your Amazon ECS cluster.

Amazon Elastic Container Service (ECS)

- **Amazon ECS controller**

The Amazon ECS scheduler is the software that manages your applications.

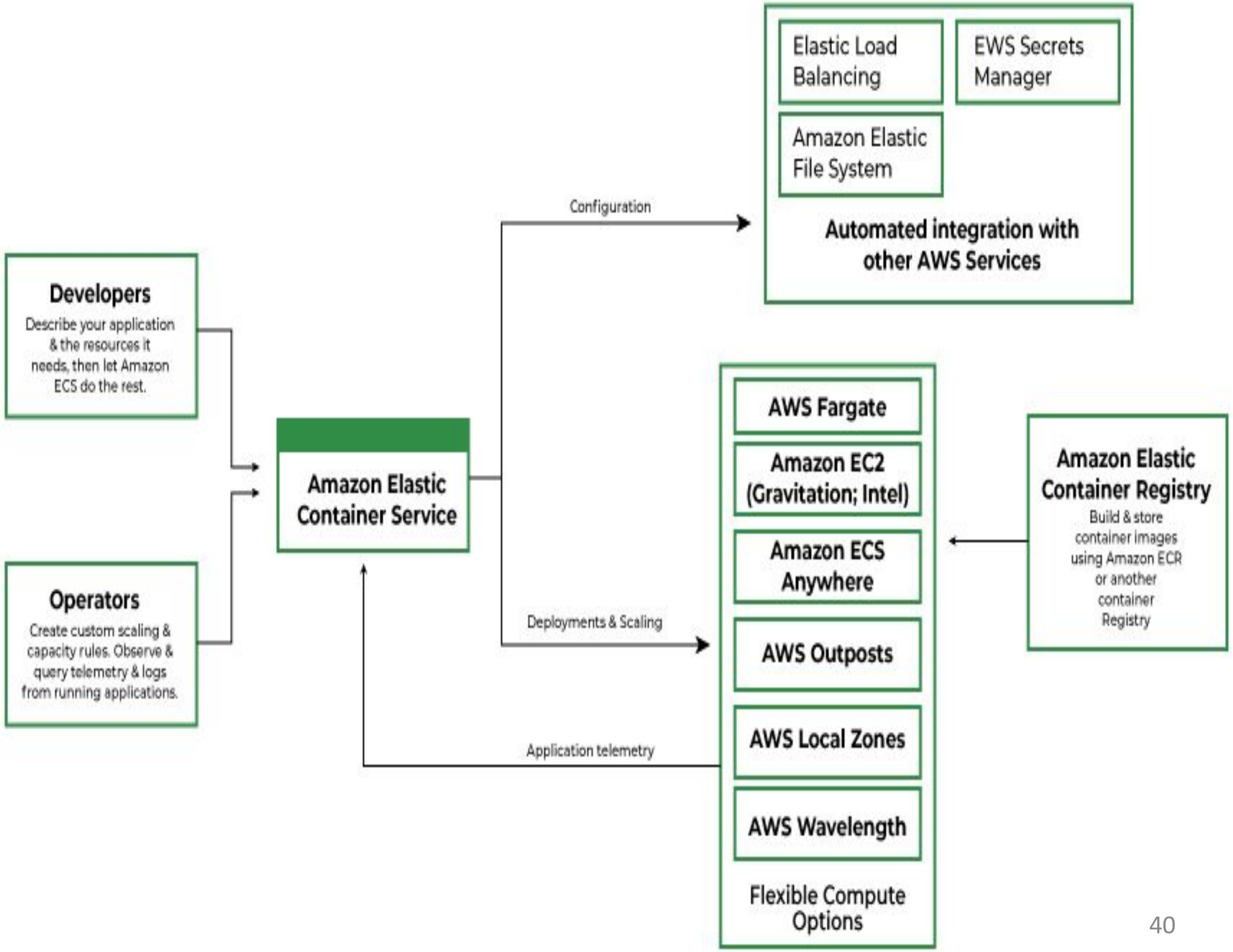
- **Amazon ECS provisioning**

There are multiple options for provisioning Amazon ECS:

- **AWS Management Console** — Provides a web interface that you can use to access your Amazon ECS resources.
- **AWS Command Line Interface (AWS CLI)** — Provides commands for a broad set of AWS services, including Amazon ECS. It's supported on Windows, Mac, and Linux.
- **AWS Software Development Kit(SDKs)** — Provides language-specific APIs and takes care of many of the connection details. These include calculating signatures, handling request retries, and error handling..
- **Copilot** — Provides an open-source tool for developers to build, release, and operate production ready containerized applications on Amazon ECS.
- **AWS Cloud Development Kit(CDK)** — Provides an open-source software development framework that you can use to model and provision your cloud application resources using familiar programming languages.

How Elastic Container Service Works?

- **Define a Task Definition** – Specify container settings (image, CPU/memory, networking).
- **Create a Cluster** – Group compute resources (EC2 or Fargate).
- **Launch a Service or Task** – Deploy a containerized application.
- **Auto-Scale & Manage** – ECS automatically scales, restarts failed tasks, and balances traffic.



How Elastic Container Service Works?

- **Container:** A container is a package that holds an application and everything dependencies, libraries, etc. the application requires to run. In other words, it is a **portable software package** that includes everything needed to run an application (Code, Runtime, Libraries, Dependencies). Containers are independent of the underlying operating system and hence container applications are fairly portable, flexible, and scalable. This ensures the application will run always as expected irrespective of the system and environment in which a container is run.
- **Docker:** Docker is an **open-source containerization platform** that allows developers to package applications with all dependencies into a single portable container. These containers can then be deployed on AWS, local servers, or any cloud platform
- **Cluster:** A logic group of EC2 instances running as a single application.
- **Container Instance:** Each EC2 in an ECS Cluster is called a container instance.

Features of ECS

Some of the features of ECS are listed below:

- Removes the need for users to manage their CPU type cluster the management system by interacting with AWS Fargate.
- Allows seamless deployment of container-based applications. This can be scheduled or done by simple API calls.
- AWS ECS takes care of the management and monitoring of the application cluster.
- Amazon ECS is region specific. This means that a cluster can only scale up/down (start-up or shut-down container instances) in a single region.
- Clusters are dynamically scalable.

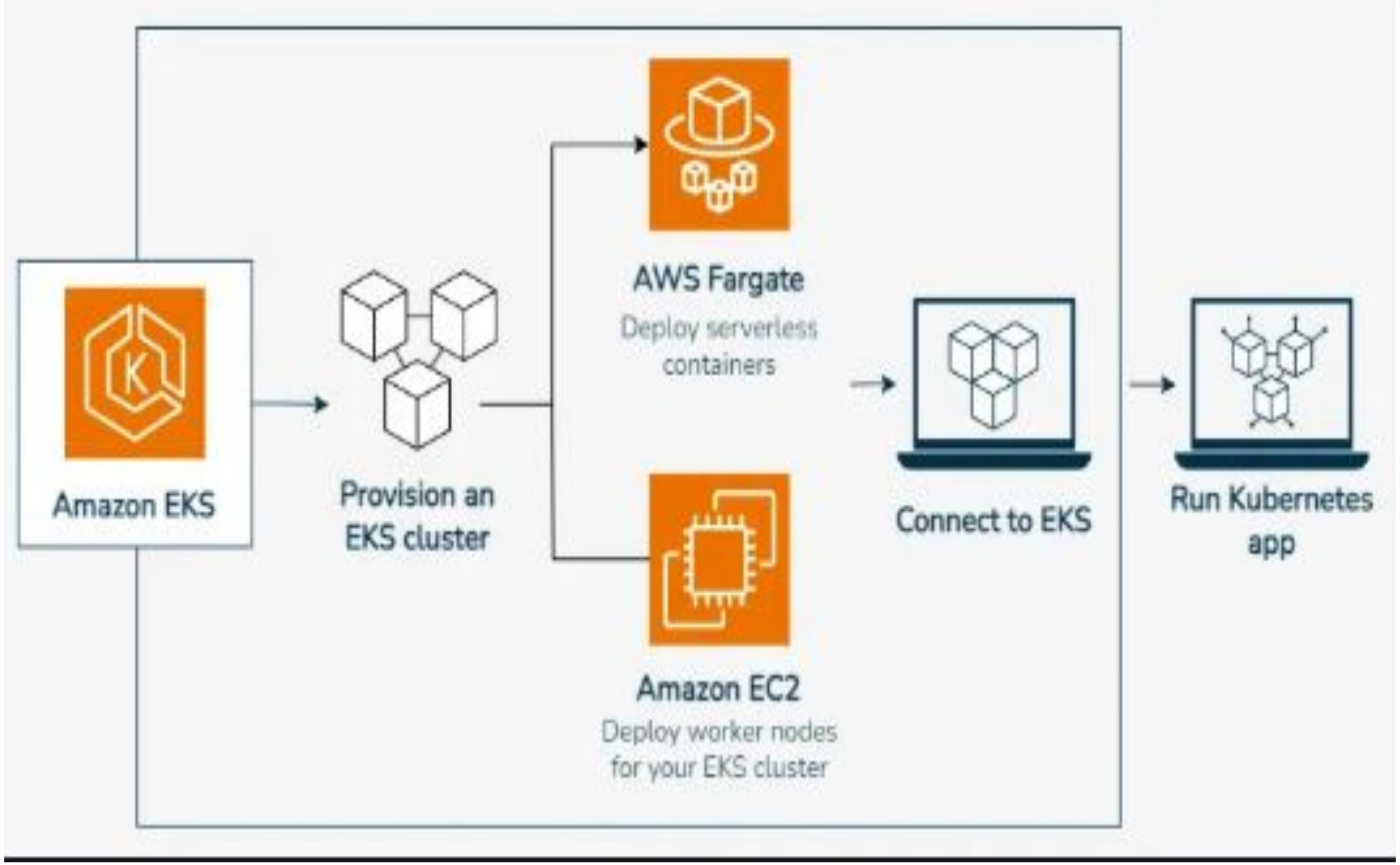
Advantages of ECS

Here are some advantages of ECS:

- **Scalability**: ECS automatically scales your applications based on demand, allowing you to easily handle changes in traffic or workload.
- **High availability**: ECS provides built-in availability and fault tolerance, ensuring that your applications are always up and running.
- **Cost-effective**: ECS enables you to optimize your infrastructure costs by scaling resources based on demand and only paying for what you use.
- **Integration**: ECS integrates with other AWS services such as Amazon ECR, AWS Fargate, Amazon CloudWatch, and AWS IAM.
- **Security**: ECS provides a secure environment to run your applications, with features such as IAM roles for tasks, VPC isolation, and encryption at rest.

Amazon Elastic Kubernetes Service(EKS)

- Amazon Elastic Kubernetes Service(EKS) is a fully managed service that you can use to run Kubernetes on Amazon Web Service.
- Kubernetes is open-source software/platform that enables you to install and manage applications at a high scale. It manages **large-scale deployments of containers** across multiple machines. **It automates the management, scaling, and deployment of containerized applications**.



Benefits of Amazon EKS

- Amazon Elastic Kubernetes Service (EKS) provides a **highly scalable and very secure way to run Kubernetes in the cloud**.
- One of the key benefits of Amazon EKS is its seamless **integration with other AWS services.** This ensures that your applications can **leverage the powerful ecosystem of AWS without complex configurations**.
- EKS also make sure of the heavy lifting in terms of managing the Kubernetes control plane which is automatically distributed across multiple availability zones. This enhances the availability and fault tolerance of your workloads.
- **Auto-scaling and scalability features** of K8s let your cluster dynamically adjust based on the demand making it an ideal solution for applications that experience variable traffic patterns
- At the End **EKS provides multi-region support** which Let you to **deploy your Kubernetes applications across different regions** for disaster recovery and high availability.
- The open-source nature of Kubernetes means that Amazon EKS is compatible with existing Kubernetes tooling making it easy to migrate workloads.

Features of AWS EKS

Following are the Features of using Amazon EKS

- **Managed Kubernetes Control Plane:** AWS automatically applies the latest patches and updates to the control plane keeping it secure and up to date with the latest Kubernetes versions. It make sure that control plane is distributed across multiple Availability Zones so that application should be highly available to us. **AWS takes care of control plane availability, scaling, and patching EKS runs across multiple AWS Availability Zones**.
- **Native Kubernetes Compatibility:** Integrating EKS with any Kubernetes-native tools such as kubectl, Helm and more easily as EKS runs **upstream Kubernetes**, which means it follows the **same open-source Kubernetes standards** as any other Kubernetes environment. Because of this, users can seamlessly **integrate and migrate workloads** from other Kubernetes clusters without significant modifications.
- **EKS Managed Node Groups:** EKS manages node updates and security patches keeping your worker nodes secure without manual intervention and can **automatically scale the number of worker nodes in response to application demands**, ensuring you have the right amount of compute power at all times.

Features of AWS EKS

- **Security and Compliance:** IAM Roles Feature allows Kubernetes service accounts to assume specific IAM roles giving you fine-grained permissions for your Kubernetes workloads.
- **Hybrid and Multi-Cloud Support:** EKS can be deployed across different regions and used in combination with other cloud providers for a true multi-cloud strategy and can use EKS on AWS Outposts to brings native AWS Services to your on-premise environments.

AWS Storage Services

- Amazon S3 is a Simple Storage Service in AWS that stores files of different types like Photos, Audio, and Videos as Objects providing more scalability and security. It is considered as an IaaS (Infrastructure as a Service).
- It allows the users to store and retrieve any amount of data at any point in time from anywhere on the web.
- It facilitates features such as extremely high availability, security, and simple connection to other AWS Services.
- Amazon S3 is used for various purposes in the Cloud because of its robust features with scaling and Securing of data.

What is Amazon S3 Used for?

The following are a few Wide Usage of Amazon S3 service.

- **Data Storage:** Amazon s3 acts as the best option for scaling both small and large storage applications. It helps in storing and retrieving the data-intensive applications as per needs in ideal time.
- **Backup and Recovery:** Many Organizations are using Amazon S3 to backup their critical data and maintain the data durability and availability for recovery needs.
- **Hosting Static Websites:** Amazon S3 facilitates in storing HTML, CSS and other web content from Users/developers allowing them for hosting Static Websites benefiting with low-latency access and cost-effectiveness.
- **Data Archiving:** Amazon S3 Glacier service integration helps as a cost-effective solution for long-term data storing which are less frequently accessed applications.

Amazon S3 Storage Classes

This storage maintains the originality of data by inspecting it. Types of storage classes are as follows:

- Amazon S3 Standard
- Amazon S3 Intelligent-Tiering
- Amazon S3 Express One Zone
- Amazon S3 Standard-Infrequent Access
- Amazon S3 One Zone-Infrequent Access
- Amazon S3 Glacier Instant Retrieval
- Amazon S3 Glacier Flexible Retrieval
- Amazon S3 Glacier Deep Archive
- Amazon S3 storage classes for AWS Dedicated Local Zones
- Amazon S3 on Outposts

Amazon S3 Storage Classes

1. **S3 Standard: The Go-to for Frequently Accessed Data**

- This is the default storage class for frequently accessed data requiring high performance and low latency. It offers high availability (99.99%) and is suitable for a wide range of use cases, including cloud applications, dynamic websites, and big data analytics
- Characteristics of S3 Standard:

- Availability criteria are quite good like 99.9%.
- General purpose storage for frequently accessed data
- Low latency and high throughput performance
- Improves the recovery of an object file.

Amazon S3 Storage Classes

2. S3 Intelligent-Tiering: Automated Cost Optimization for Data with Unknown Access Patterns:

- It is a smart file cabinet that automatically moves your files to the cheapest storage location without you having to do anything. It monitors how often you access your files and moves them between different storage "tiers" – from a fast, expensive one for frequently used files to a slow, cheap one for files you rarely touch.

Amazon S3 Storage Classes

2. S3 Intelligent-Tiering: Automated Cost Optimization for Data with Unknown Access Patterns:

- The first cloud storage automatically decreases the user's storage cost. It provides very cost-effective access based on frequency, without affecting other performances.
- **It is the only cloud storage class that delivers automatic storage cost savings when data access patterns change, without performance impact or operational overhead. The Amazon S3 Intelligent-Tiering storage class is designed to optimize storage costs by automatically moving data to the most cost-effective access tier when access patterns change.**
- S3 Intelligent-Tiering is the ideal storage class for data with unknown, changing, or unpredictable access patterns, independent of object size or retention period. You can use S3 Intelligent-Tiering as the default storage class for virtually any workload, especially data lakes, data analytics, new applications, and user-generated content. There are no retrieval charges in S3 Intelligent-Tiering.

- Characteristics of S3 Intelligent-Tiering:

- Required less monitoring and automatically tier charge.
- Small monthly monitoring and automation charge.
- No operational overhead, no lifecycle charges, no retrieval charges, and no minimum storage duration.
- Automatic cost savings for data with unknown or changing access patterns

Amazon S3 Storage Classes

3. Amazon S3 Express One Zone:

- Amazon S3 Express One Zone is a super-fast storage service for applications that need instant access to their data, like those for machine learning or analytics. It's designed to be 10x faster than standard S3, with speeds up to 10x faster and up to 80% lower request costs, because it stores your data in a single location (an availability zone) for lower latency.

Amazon S3 Storage Classes

3. Amazon S3 Express One Zone:

- Different from other S3 Storage Classes which store data in a minimum of three Availability Zones, S3 One Zone-IA stores data in a single Availability Zone.
- S3 Express One Zone can improve data access speeds by 10x and reduce request costs by 50% compared to S3 Standard.
- While you have always been able to choose a specific AWS Region to store your S3 data, with S3 Express One Zone you can select a specific AWS Availability Zone within an AWS Region to store your data.
- It’s a very good choice for storing secondary backup copies of on-premises data or easily re-creatable data. S3 One Zone-IA provides you the same high durability, high throughput, and low latency as in S3 Standard.

- **Characteristics :**

- Availability Zone destruction can damage the data.
- Availability is 99.5% in S3 one Zone- Infrequent Access.
- Durability is of 99.999999999%.
- High performance storage for your most frequently accessed data
- Optimized for large datasets with many small objects
- Scale to handle millions of requests per minute

Amazon S3 Storage Classes

4. S3 Standard-Infrequent Access: Cost-Effective Storage for Less Frequently Used Data

- To access the less frequently used data, users use S3 Standard-IA. It requires rapid access when needed. We can achieve high strength, high output, and low bandwidth by using S3 Standard-IA. It is best in storing the backup, and recovery of data for a long time. It act as a data store for disaster recovery files.
- **Standard-IA is for data that is accessed less frequently, but requires rapid access when needed. S3 Standard-IA offers the high durability, high throughput, and low latency**
- Characteristics of S3 Standard-Infrequent Access

- High performance and same action rate.
- Infrequently accessed data that needs millisecond access
- Same low latency and high throughput performance of S3 Standard
- Very Durable in all AZs.
- Availability is 99.9% in S3 Standard-IA.
- Durability is of 99.999999999%

Amazon S3 Storage Classes

5. Amazon S3 One Zone-Infrequent Access: Cost-Effective Storage for Less Frequently Used Data

- Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) is a storage option for data you don't access often but need to retrieve quickly when you do. It's cheaper than standard storage because it only stores your data in one data center (an Availability Zone) instead of the three used by other classes. It's perfect for things like secondary backups or data that is easy to recreate.

Key features in simple terms

- Less expensive: By storing data in only one location, it costs less than other S3 options that store data across multiple locations.
- Quick access: When you need the data, you can still access it quickly, just like with standard storage.
- Single location: Your data is not duplicated across multiple data centers. If that single data center has a problem, your data could be lost.

Amazon S3 Storage Classes

5. Amazon S3 One Zone-Infrequent Access: Cost-Effective Storage for Less Frequently Used Data

- S3 One Zone-IA is for data that is accessed less frequently, but requires rapid access when needed. Unlike other S3 Storage Classes which store data in a minimum of three Availability Zones (AZs), S3 One Zone-IA stores data in a single AZ and costs 20% less than S3 Standard-IA.
- It’s a good choice for storing secondary backup copies of on-premises data or easily re-creatable data.
- S3 One Zone-IA is ideal for customers who want a lower-cost option for infrequently accessed data
- Characteristics:

 Re-creatable infrequently accessed data

 Same low latency and high throughput performance of S3 Standard

 Availability is 99.9% in S3 Standard-IA.

 Durability is of 99.999999999%

Amazon S3 Storage Classes

6. S3 Glacier Instant Retrieval: High-Performance Archiving with Rapid Retrieval

- S3 Glacier Instant Retrieval is like a low-cost, long-term storage locker for digital files that you don't need to access often, but when you do, you need it immediately, like a photo or medical scan. It's cheaper than frequently accessed storage, but unlike other long-term archives, it delivers your files back in milliseconds instead of hours.

How it works

- For long-lived, rarely accessed data: Think of things like user-generated content, media archives, or medical records that need to be kept for years but are only viewed infrequently.
- Instant access when you need it: When a user or application needs the file, it's retrieved in milliseconds, just as fast as if it were in a standard, more expensive storage system.
- Significant cost savings: You can save up to 68% on storage costs compared to storing that data on a standard "infrequent access" plan, especially if you only access the files about once a quarter.

Amazon S3 Storage Classes

6. S3 Glacier Instant Retrieval: High-Performance Archiving with Rapid Retrieval

- It is an archive storage class that delivers the lowest-cost storage for data archiving and is organized to provide you with the highest performance and with more flexibility. S3 Glacier Instant Retrieval delivers the fastest access to archive storage. Same as in S3 standard, Data retrieval in milliseconds .

- Characteristics of S3 Glacier Instant Retrieval

 It just takes milliseconds to recover the data.

 The minimum object size should be 128KB.

 Availability is 99.9% in S3 glacier Instant Retrieval.

 Durability is of 99.999999999%

 Long-lived data that is accessed a few times per year with instant retrievals

Amazon S3 Storage Classes

7.S3 Glacier Flexible Retrieval: Balancing Cost and Retrieval Flexibility for Archiving

- S3 Glacier Flexible Retrieval is a very low-cost way to store data that you don't need to access immediately, like backups or long-term archives. You can retrieve this data within minutes to hours, depending on the option you choose, and it's ideal for scenarios like disaster recovery where data might not be needed for a long time.

How it works

- Low cost: It's significantly cheaper than standard online storage for data that you rarely access (e.g., once or twice a year).
- Retrieval flexibility: It offers several options for getting your data back, balancing cost and retrieval time.
- Expedited: 1 to 5 minutes.
- Standard: 3 to 5 hours.
- Bulk: 5 to 12 hours.
- Free bulk retrievals: You can retrieve large amounts of data in bulk for free, which is great for backups.
- Asynchronous access: Unlike standard storage, you have to request a restore before you can access the data. The data is not instantly available.

Amazon S3 Storage Classes

7.S3 Glacier Flexible Retrieval: Balancing Cost and Retrieval Flexibility for Archiving

- S3 Glacier Flexible Retrieval delivers low-cost storage, up to 10% lower cost (than S3 Glacier Instant Retrieval), for archive data that is accessed 1—2 times per year and is retrieved asynchronously.
- For archive data that does not require immediate access but needs the flexibility to retrieve large sets of data at no cost, such as backup or disaster recovery use cases, S3 Glacier Flexible Retrieval (formerly S3 Glacier) is the ideal storage class.
- It provides low-cost storage compared to S3 Glacier Instant Retrieval. It is a suitable solution for backing up the data so that it can be recovered easily a few times in a year. It just takes minutes to access the data.

- Characteristics :

- Free recoveries in high quantity.
- Backup and archive data that is rarely accessed and low cost
- Supports SSL for data in transit and encryption of data at rest
- When you have to retrieve large data sets , then S3 glacier flexible retrieval is best for backup and disaster recovery use cases.
- Availability is 99.99% in S3 glacier flexible retrieval.
- Durability is of 99.999999999%.

Amazon S3 Storage Classes

8. Amazon S3 Glacier Deep Archive

- Amazon S3 Glacier Deep Archive is like a digital, super-cheap "long-term storage locker" for data you rarely need, such as legal records or old backups. It's the lowest-cost option for keeping data safe for years, but if you need to get to it, it will take several hours to retrieve. It's an easy and affordable alternative to managing old physical tapes.

In simple terms:

- It's for rarely accessed data: Think of data that you might only look at once a year, or even less.
- It's extremely cheap: It costs even less than other online storage options, making it great for storing large amounts of data for a long time.
- Retrieval takes time: Unlike a file you access on your computer, getting data out of Deep Archive takes time—typically 12 hours, or up to 48 hours for large amounts.
- It's very safe: The data is stored redundantly across multiple physical locations to protect it from being lost, giving it very high durability.
- It's a replacement for tape: It eliminates the need for expensive and difficult-to-manage physical tape storage.

Amazon S3 Storage Classes

8. Amazon S3 Glacier Deep Archive

- The Glacier Deep Archive storage class is designed to provide long-lasting and secure long-term storage for large amounts of data at a price that is competitive with off-premises tape archival services that is very cheap.
- It is designed for customers—particularly those in highly-regulated industries, such as financial services, healthcare, and public sectors—that retain data sets for 7—10 years or longer to meet regulatory compliance requirements.
- You no longer need to deal with expensive services. Accessibility is very much efficient, that it can restore data within 12 hours.
- This storage class is designed in such a way that users can easily get long-lasting and more secured storage for a huge amount of data at very less cost. Efficient accessibility and can restore data within very less time, therefore its time complexity is also efficient. S3 Glacier Deep Archive also have the feature of objects replication.

- Characteristics of S3 Glacier Deep Archive

- More secured storage.
- Archive data that is very rarely accessed and very low cost
- Retrieval time within 12 hours
- Availability is 99.99% in S3 glacier deep archive.
- Durability is of 99.999999999%.

Amazon S3 Storage Classes

9. Amazon S3 storage classes for AWS Dedicated Local Zones

- For AWS Dedicated Local Zones, the S3 storage classes (S3 Express One Zone and S3 One Zone-Infrequent Access) are specific options for storing data within that private, physical location. They are designed for use cases that require data to be in a specific, isolated area to meet data residency or isolation rules, which can't be met by a regular AWS Region. These classes provide high performance for frequently accessed data or low-cost storage for less frequently accessed data, all while keeping the data within the boundaries of the Dedicated Local Zone.

- Characteristics:

 Store S3 objects in a specific data perimeter

 Designed to durably and redundantly store data in a single Dedicated Local Zone

Amazon S3 Storage Classes

9. Amazon S3 storage classes for AWS Dedicated Local Zones

- In AWS Dedicated Local Zones, the S3 Express One Zone and S3 One Zone-Infrequent access storage classes are purpose-built to store data in a specific data perimeter to support your data isolation and data residency use cases.
- Dedicated Local Zones are a type of AWS infrastructure that is fully managed by AWS, built for exclusive use by you or your community, and placed in a location or data center specified by you to help you comply with regulatory requirements. Both storage classes store data in a single Dedicated Local Zone and are supported in directory buckets.

- Characteristics:

 Store S3 objects in a specific data perimeter

 Designed to durably and redundantly store data in a single Dedicated Local Zone

Amazon S3 Storage Classes

10. Amazon S3 on Outposts

- Amazon S3 on Outposts is a service that lets you use Amazon's S3 object storage directly in your own data center, using the same familiar S3 commands and features. It's designed for applications that need to store and process data locally because of latency, data residency, or performance requirements, by extending the AWS cloud to your on-premises facility.

In simple terms

- What it is: Imagine you have a large amount of data on your company's computers that needs to be stored in a very organized way, like a massive online filing cabinet. Amazon S3 is AWS's service for doing this in the cloud.

- What it does: Amazon S3 on Outposts is the same filing cabinet service, but it's installed directly in your office or factory using AWS hardware (called an Outpost).

Amazon S3 Storage Classes

10. Amazon S3 on Outposts

- Amazon S3 on Outposts delivers object storage to your on-premises AWS Outposts environment. Using the S3 APIs and features available in AWS Regions today, S3 on Outposts makes it easy to store and retrieve data on your Outpost, as well as secure the data, control access, tag, and report on it.
- An Outpost is a pool of AWS compute and storage capacity deployed at a customer site. AWS Outposts is a fully managed service that extends AWS infrastructure, services, APIs, and tools to customer premises. By providing local access to AWS managed infrastructure, AWS Outposts enables customers to build and run applications on premises using the same programming interfaces as in AWS Regions, while using local compute and storage resources for lower latency and local data processing needs.
- The S3 Outposts storage class is ideal for workloads with local data residency requirements, and to satisfy demanding performance needs by keeping data close to on-premises applications(On-premises in AWS refers to physical devices).
- Characteristics:

 Store S3 objects in your on-premises AWS Outposts environment

 Designed to durably and redundantly store data on your AWS Outposts rack

 Authentication and authorization using IAM and S3 Access Points

Amazon S3 Storage Classes

S3 Standard	S3 Intelligent-Tiering	S3 Standard-IA	S3 One Zone-IA	S3 Glacier	S3 GlacierDeep Archive
Frequent		Access frequency			Archive
• Active, frequently accessed data • Milliseconds access • > 3 AZ	• Data with changing access patterns • Milliseconds access • > 3 AZ • Monitoring fee per object • Min storage duration	• Infrequently accessed data • Milliseconds access • > 3 AZ • Retrieval fee per GB • Min storage duration • Min object size	• Re-creatable, less accessed data • Milliseconds access • 1 AZ • Retrieval fee per GB • Min storage duration • Min object size	• Archive data • Select minutes or hours • > 3 AZ • Retrieval fee per GB • Min storage duration • Min object size	• Long-term archive data • Select hours • > 3 AZ • Retrieval fee per GB • Min storage duration • Min object size

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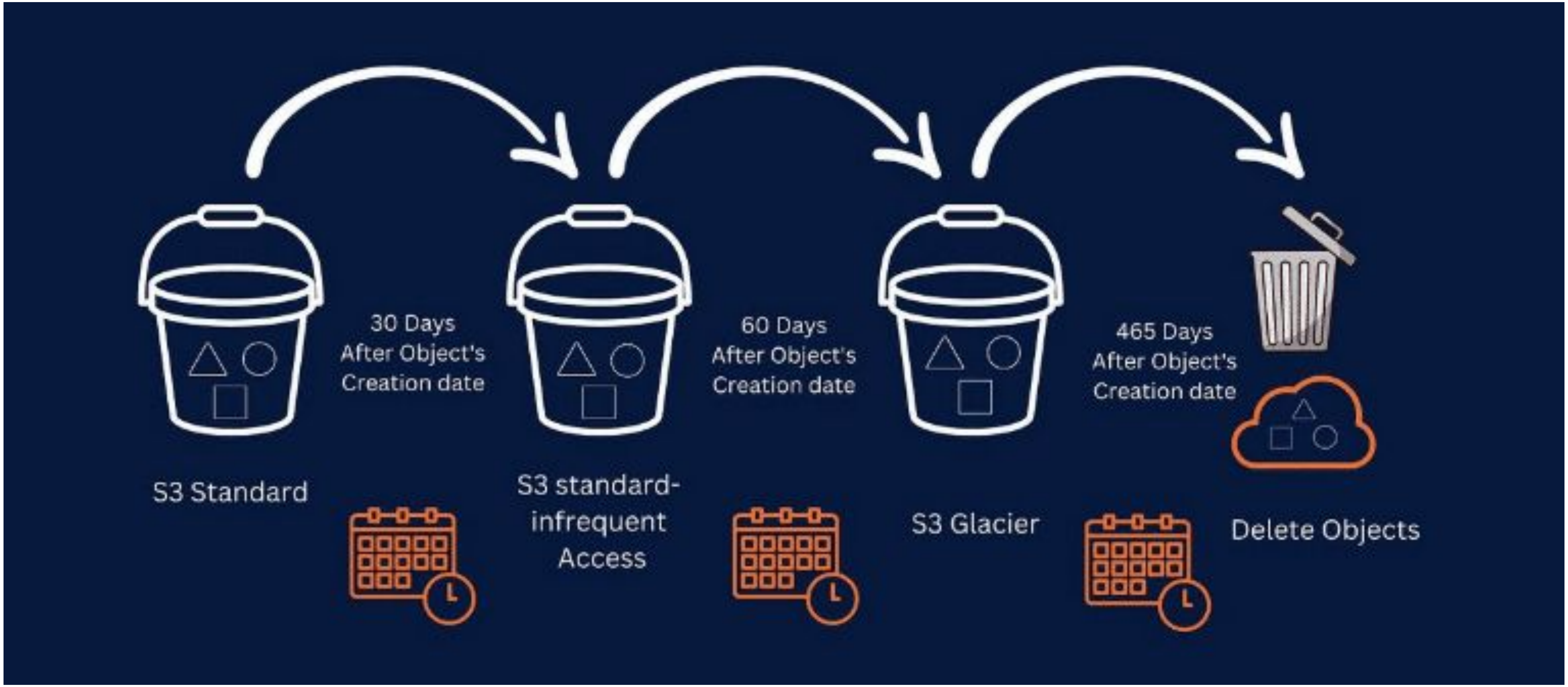
S3 Storage Class	Simple Term	When to Use It	Key Trade-off
S3 Standard	"The Everyday Files"	For data you use all the time, like website content or active application files, that needs immediate access.	Most expensive for storage, but no retrieval fees.
S3 Standard-Infrequent Access (S3 Standard-IA)	"The Backups Box"	For important data you don't access often (like backups or old reports), but when you do need it, you need it immediately.	Lower storage cost than Standard, but you pay a fee to get the data back (retrieval fee).
S3 One Zone-Infrequent Access (S3 One Zone-IA)	"The Risky Backups Box"	Like the "Backups Box," but the data is stored in only one physical location. Use this only if you can easily re-create the data if that location is hit by a disaster (like a fire or flood).	Even lower storage cost than Standard-IA, but less protection against a regional disaster.
S3 Intelligent-Tiering	"The Smart Mover"	For data where you have no idea how often it will be accessed. AWS automatically moves it between "everyday" and "infrequent access" tiers to save you money without you having to do anything.	A small monitoring fee, but no retrieval fees and no manual management needed.
S3 Glacier Instant Retrieval	"The Instant Archive"	For archive data you rarely touch (a few times a year), but if you do need it, you need it instantly (milliseconds retrieval).	Very low storage cost, but retrieval fees apply. Minimum storage duration of 90 days.
S3 Glacier Flexible Retrieval	"The Archive Warehouse"	For data you access very rarely, and you can wait a few minutes to several hours to get it back. Ideal for general archiving and disaster recovery.	Extremely low storage cost, but retrieval takes time. Minimum storage duration of 90 days.
S3 Glacier Deep Archive	"The Deep Freeze Vault"	The cheapest option for data you plan to store for years (7-10 years or more) and almost never need (once or twice a year). Retrieval can take 12 to 48 hours.	The lowest possible storage cost, but slowest retrieval time. Minimum storage duration of 180 days.

S3 Lifecycle Policy



S3 Lifecycle Policy

- An S3 Lifecycle Policy is a set of automated rules that tell Amazon S3 what to do with your stored files (objects) over time. Think of it like a smart assistant for your S3 bucket that helps you manage your data and save money.

Here's how it works in simple terms:

- Transition Actions (Moving Files): You can set rules to automatically move files between different S3 storage classes.
- For example: "After 30 days, move these frequently accessed files to a cheaper 'Infrequent Access' storage class because I don't use them as often anymore."
- "After 90 days, move these old log files to 'Glacier' or 'Glacier Deep Archive' for long-term, very cheap storage, as I only need them for compliance."

Benefits of Implementing Lifecycle Policies

Implementing S3 lifecycle policies offers several benefits to businesses:

- **Cost optimization:** By automatically moving less frequently accessed data to lower-cost storage classes, businesses can significantly reduce their storage costs without manual intervention.
- **Efficient data management:** Lifecycle policies streamline data management by automating the process of transitioning objects between storage classes or deleting them, saving time and effort for IT teams.
- **Compliance and security:** Lifecycle policies can help enforce data retention policies, ensuring compliance with regulatory requirements. They also enhance security by automating the deletion of sensitive data after its required retention period, reducing the risk of data breaches.
- **Improved performance:** By keeping the storage optimized and clutter-free, lifecycle policies contribute to improved system performance, enabling faster data access and retrieval times.

S3 Lifecycle Policy Examples

Here are some common S3 lifecycle policy examples:

- **Deleting Objects**

One common use of S3 lifecycle policies is to automatically delete objects after a specified period. For instance, businesses can set up a policy to delete temporary files or logs older than 90 days, ensuring that unnecessary data does not clutter their storage space indefinitely.

- **Moving Objects**

Lifecycle policies can also be configured to move objects between different storage classes. For example, you can transition frequently accessed data to the S3 Standard storage class for optimal performance and move less frequently accessed data to the S3 Intelligent-Tiering class to save costs without sacrificing availability and latency.

Amazon EBS (Elastic Block Store)

- It is an Infrastructure as a Service (IaaS) component that provides persistent block storage for EC2 instances. It functions like a virtual hard drive that you can attach to an EC2 instance to store data persistently, even if the instance is stopped.

- AWS EBS is like a virtual hard drive for your cloud-based servers (EC2 instances). It provides high-performance, persistent storage that you can attach to and detach from these instances as needed, just like a physical hard drive. Think of it as a reliable and scalable way to store data, run databases, or install operating systems on your cloud servers.

- EBS is designed specifically to be used exclusively with Amazon EC2 instances. It acts as a persistent, block-level storage volume that you attach to an EC2 instance to store data, and a volume cannot be used until it is attached to an instance.

Amazon EBS Vs S3

- AWS EBS is like a virtual hard drive for your cloud-based servers (EC2 instances).
- It provides high-performance, persistent storage that you can attach to and detach from these instances as needed, just like a physical hard drive.
- Think of it as a reliable and scalable way to store data, run databases, or install operating systems on your cloud servers.

Amazon EBS Vs S3

Feature	Amazon EBS	Amazon S3
Analogy	A hard drive for one computer	A vast storage room
Data Type	Block storage (raw data)	Object storage (files, images, videos)
Primary Use	Boot volumes, databases, and transactional applications that require low-latency access	Backups, media storage, static website content, and big data analytics
Access	Directly attached to a single $(EC2)$ instance	Accessible over the internet via a unique identifier
Scalability	Limited to a certain size (up to 64 TB)	Virtually unlimited storage
Performance	Optimized for high-performance, low-latency access	Optimized for high throughput and scalability; higher latency than EBS

Amazon EBS (Elastic Block Store)

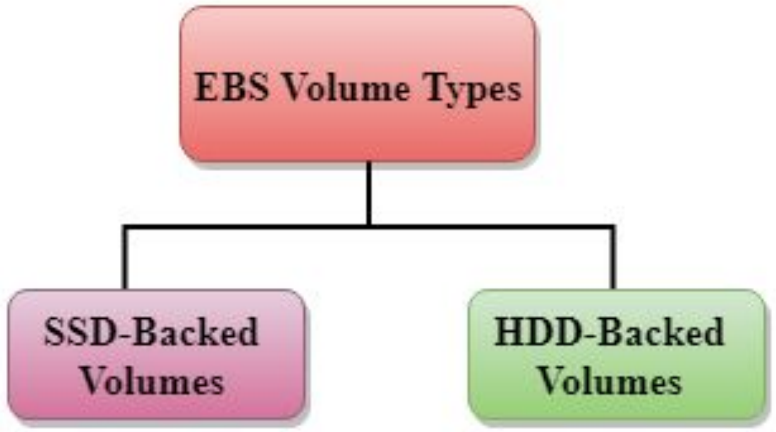
- EC2 is a virtual server in a cloud while EBS is a virtual disk in a cloud.
- Amazon EBS allows you to create storage volumes and attach them to the EC2 instances.
- Once the storage volume is created, you can create a file system on the top of these volumes, and then you can run a database, store the files, applications or you can even use them as a block device in some other way.
- Amazon EBS volumes are placed in a specific availability zone, and they are automatically replicated to protect you from the failure of a single component.
- EBS volume does not exist on one disk, it spreads across the Availability Zone. EBS volume is a disk which is attached to an EC2 instance.
- EBS volume attached to the EC2 instance where windows or Linux is installed known as Root device of volume.

EBS Volume Types

Amazon EBS provides two types of volume that differ in performance characteristics and price. EBS Volume types fall into two parts:

1. SSD-backed volumes

2. HDD-backed volumes



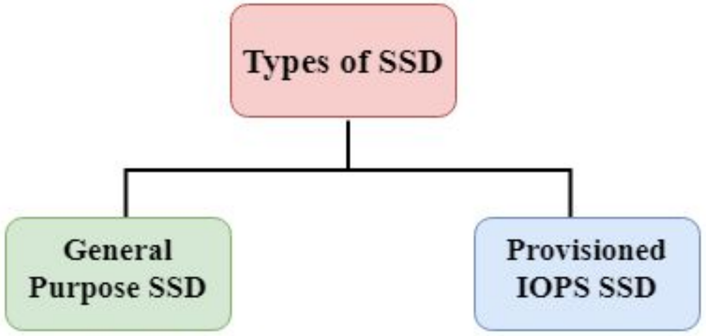
1. SSD

- SSD stands for solid-state Drives.
- In June 2014, SSD storage was introduced.
- It is a general-purpose storage.
- It supports up to 4000 IOPS which is quite very high.
- SSD storage is very high performing, but it is quite expensive as compared to HDD (Hard Disk Drive) storage.

EBS Volume Types

SSD is further classified into two parts:

- General Purpose SSD
- Provisioned IOPS(Input/Output Operations Per Second) SSD



General Purpose SSD

- General Purpose SSD is also sometimes referred to as a GP2.
- It is a General-purpose SSD volume that balances both price and performance.

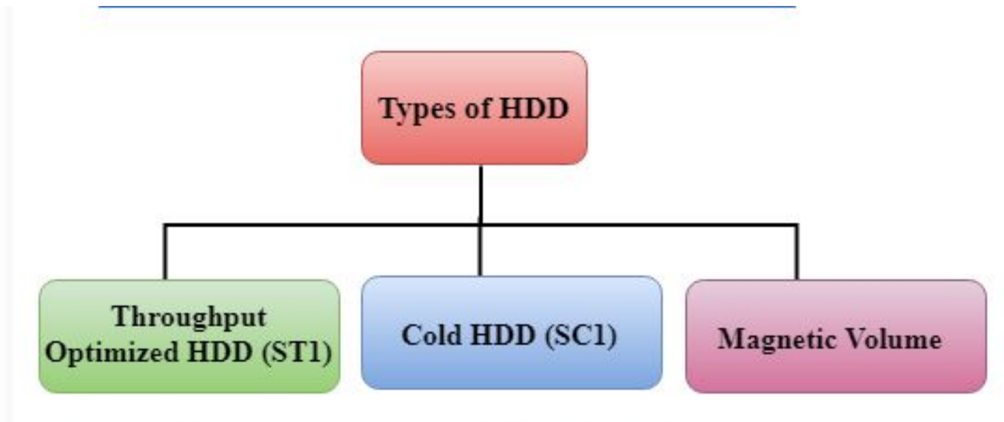
Provisioned IOPS SSD

- It is also referred to as IO1.
- High-performance storage for mission-critical, performance-sensitive applications like large transactional databases.
- It is mainly used for high-performance applications such as intense applications, relational databases.
- It is designed for I/O intensive applications such as large relational or NOSQL databases.
- It is used when you require more than 10,000 IOPS
- General Purpose SSD (gp2) volumes are cost-effective and balance performance and price, while Provisioned IOPS SSD (io1) volumes offer the highest IOPS performance and are best for demanding applications

EBS Volume Types

2. HDD

- It stands for Hard Disk Drive.
- HDD based storage was introduced in 2008.
- The size of the HDD based storage could be between 1 GB to 1TB.
- It can support up to 100 IOPS which is very low.



EBS Volume Types

Throughput Optimized HDD (st1)

- It is also referred to as ST1, which is the specific volume type identifier used by Amazon Web Services (AWS).
- Throughput Optimized HDD is a low-cost HDD designed for those applications that require higher throughput up to 500 MB/s.
- It is useful for those applications that require the data to be frequently accessed.
- It is used for Big data, Data warehouses, etc.
- It cannot be a boot volume, so it contains some additional volume. For example, if we have Windows server installed in a C: drive, then C drive cannot be a Throughput Optimized Hard disk, D: drive or some other drive could be a Throughput Optimized Hard disk.(A **boot volume** is the primary disk that contains the **operating system (OS)** and is required for the system to start (boot). **AWS does not allow st1 volumes to be used as boot volumes** because these disks are optimized for large sequential data transfers, **not for fast random access**, which is needed for OS operations. Instead, the **boot volume must be a General Purpose SSD (gp2/gp3) or Provisioned IOPS SSD (io1/io2).)**
- Ex: When launching an **EC2 instance**, you cannot choose an **st1 volume for the root/boot drive**. Instead, you use an **SSD (gp3 or io2)** for booting and add **st1 for extra storage**.
- The size of the Throughput Hard disk can be 500 GiB to 16 TiB.

EBS Volume Types

Cold HDD (sc1)

- It is also known as SC1.
- It is the lowest cost storage designed for the applications where the workloads are infrequently accessed.
- It is useful when data is rarely accessed.
- It is mainly used for a File server.
- It can’t be a boot volume.
- The size of the Cold Hard disk can be 500 GiB to 16 TiB.
- It supports up to 250 IOPS.
- **Cold HDD (sc1)** is a type of **Amazon Elastic Block Store (EBS) volume** that provides **low-cost, high-throughput storage** for workloads that access data infrequently. It is the **cheapest EBS storage option** and is best suited for **cold data storage** needs.

EBS Volume Types

Magnetic Volume

- It is the lowest cost storage per gigabyte of all EBS volume types.
- AWS Magnetic Volumes are a cost-effective storage type for frequently accessed data
- It is useful for applications where the lowest storage cost is important.
- Magnetic volume is the only hard disk which is bootable. Therefore, we can say that it can be used as a boot volume.
- **Magnetic Volume** was one of the earlier **Amazon Elastic Block Store (EBS) volume types**, providing **low-cost storage** for applications that require **infrequent access to data**. Unlike **Cold HDD (sc1)** or **Throughput Optimized HDD (st1)**, Magnetic volumes were **bootable**, making them unique among AWS's HDD-based storage options.

Amazon Glacier

- Amazon Glacier, also known as Amazon Simple Storage Service (S3) Glacier, is a low-cost cloud storage service for data with longer retrieval times offered by Amazon Web Services (AWS).

- Amazon Glacier provides storage for data archiving and backup of cold data. Cold data refers to files that are infrequently accessed but are kept in case they are needed at a later date.

- A developer will use a cold data service such as Amazon Glacier to move data that is not needed often to archival storage to save money on storage costs.

- Amazon advertises Amazon Glacier as an extremely low-cost storage service. To keep costs low, the service is optimized for infrequently accessed data -- where data retrieval times can reach from three to five hours.

- Amazon S3 Glacier is a **low-cost cloud storage service** provided by **AWS** for **data archiving and backup of cold data**.
- **Cold data** refers to **infrequently accessed files** that must be **retained for long periods** (e.g., backups, logs, compliance records).
- Amazon Glacier **reduces storage costs** by optimizing for **longer data retrieval times**.
- Data retrieval can take **several hours (3 to 5 hours)** instead of being **instantaneous** like S3 Standard.

Amazon Glacier

- Amazon Glacier is an extremely low-cost cloud storage service designed for data you need to keep for a long time but rarely, if ever, access. It's like putting documents into deep, secure storage (like an off-site archive or safety deposit box) where they are safe for years, but you don't need instant access to them.
- When an organization wants to retrieve data, it has three options:

- **Bulk retrievals**, which recover large amounts of data in a five- to 12-hour window; this is the **lowest cost option**.

- Standard retrievals, which take about three to five hours to recover stored data.

- **Expedited retrievals**, which can return data **within five minutes**. This option, which is the **most costly**, is designed for organizations that may need their stored data at any time.

- Additionally, Amazon considers Amazon Glacier an official part of Amazon Simple Storage Service, which is why it is also called S3 Glacier.

What are the benefits of Amazon Glacier?

Amazon Glacier offers the following benefits:

- **Lower cost.** Glacier is designed as Amazon's lowest-cost storage class. This enables an organization to store large amounts of data at a reduced cost compared to other Amazon storage services.
- **Maintains archival database.** An organization does not have to maintain its own archival database. AWS handles administrative tasks such as capacity planning and hardware
- **Durability.** Glacier is distributed across at least three physical AWS Availability Zones at a time, increasing the ability to restore data if it is lost in one zone.
- **Scalability.** Organizations can scale the stored data up or down as needed.
- **Multiple data retrieval options.** Organizations can choose from expedited, standard and bulk retrievals.
- **Security.** S3 Glacier supports security and compliance standards such as General Data Protection Regulation, Payment Card Industry Data Security Standard, Health Insurance Portability and Accountability Act and Federal Information Security Management Act. It also supports encryption and monitors storage application programming interface call activities.
- **Integration**. Glacier integrates with other AWS offerings such as AWS Snowball and AWS Direct Connect.

Glacier vs. Amazon S3

Amazon S3 is like a general-purpose cloud hard drive for data you need to access quickly and frequently, while Amazon S3 Glacier is like a cheap, long-term archive box for data you rarely need but want to keep safe and low-cost. Although Amazon considers Glacier an official part of S3, they are still two separate storage options with two different use cases.

- The biggest difference between the two Amazon storage services is that S3 is designed for data that needs to be retrieved in real time, while Amazon Glacier is used for archival.
- Use of S3 Glacier is reserved for low storage cost use cases where data will not be needed at a moment's notice. S3, however, is suggested for when an organization needs frequent and fast access to its data.
- While S3 Glacier uses archives and vaults to store data, S3 uses storage buckets.

Glacier vs. Amazon S3

- Amazon S3 is a durable, secure, simple, and fast storage service, while Amazon S3 Glacier is used for archiving solutions.
- Use S3 if you need low latency or frequent access to your data. Use S3 Glacier for low storage cost, and you do not require millisecond access to your data.
- You have three retrieval options when it comes to Glacier, each varying in the cost and speed it retrieves an object for you. You retrieve data in milliseconds from S3.
- Both S3 and Glacier are designed for durability of 99.999999999% of objects across multiple Availability Zones.
- S3 and Glacier are designed for availability of 99.99%.
- S3 can be used to host static web content, while Glacier cannot.

Amazon Elastic File System(EFS)

- Amazon EFS is like a shared, central hard drive in the cloud that many computers can use at the same time to store and access files.
- It's a fully managed service that automatically scales storage up or down as needed, meaning you don't have to manually provision space.
- It is ideal for applications that need to share data, such as web serving, big data analytics, and content management.

Amazon Elastic File System(EFS)

- EFS is a file-level, fully managed, storage provided by AWS (Amazon Web Services) that can be accessed by multiple EC2 instances concurrently.
- Just like the AWS EBS, EFS is specially designed for high throughput and low latency applications.
- An EFS can be attached to multiple Amazon Web Services (AWS) compute instances and on-premises servers, providing a common resource for applications and data storage in many different environments.
- Amazon states that a single EFS can be simultaneously connected to thousands of Elastic Compute Cloud (EC2) instances or on-premise resources, allowing you to share EFS data with as many resources as needed.
- Access to shared EFS folders and data is provided through native operating system interfaces.

Amazon Elastic File System(EFS)

Advantages:

- An Amazon EFS is elastic. That means its storage capacity can be automatically scaled up (add more storage) or scaled down (shrink storage capacity) as folders and files are added to or removed from the system. This is a major advantage over traditional storage solutions—you can add or remove capacity without disrupting users or applications.
- Importantly, EFS storage is permanent. When attached to an AWS compute instance, data will not disappear when that instance is relaunched.

Amazon Elastic File System(EFS)

Disadvantages:

Amazon EFSs do have a couple limitations:

- **No Windows instances.** Amazon EFSs are not supported on AWS Windows EC2 instances. EFS volumes can only be used with non-Windows instances, such as Linux, that support NFS(Network File System) volumes.

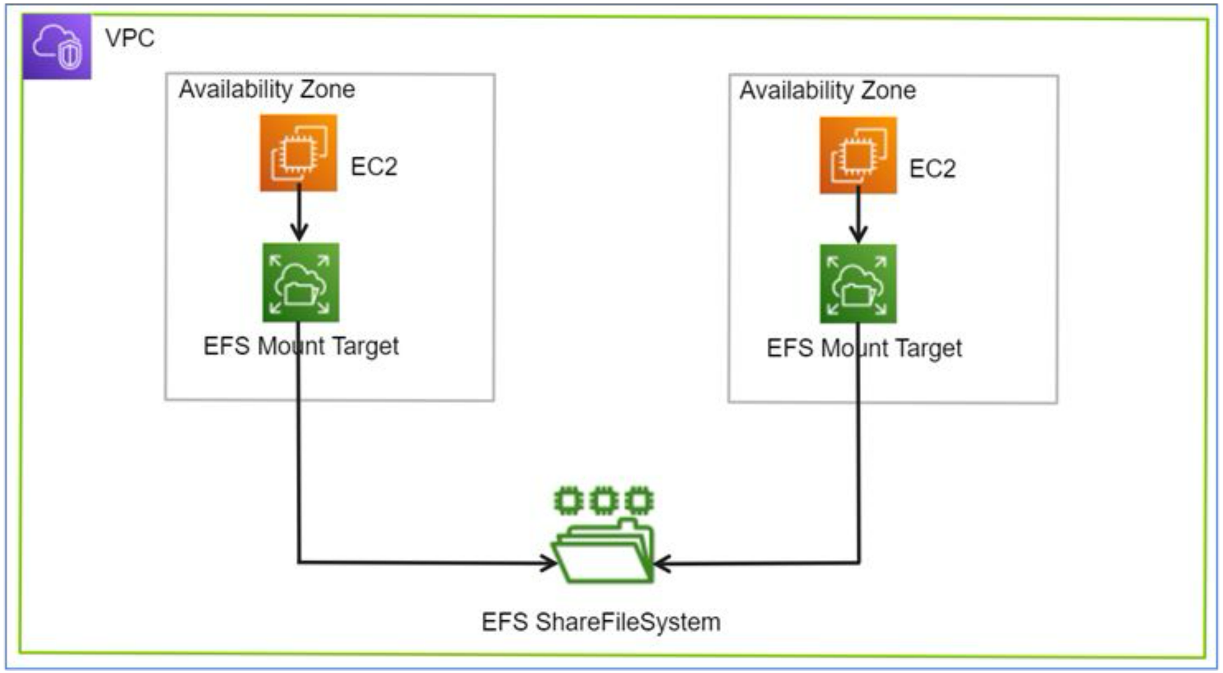
- **No system boot volumes.** Amazon EFS volumes also cannot be used for system boot volumes. AWS EC2 instances must use Elastic Block Store (EBS) volumes for booting their systems. EBS volumes are like EFS volumes with one exception. An EBS volume can only be connected to one EC2 instance or server, while EFS volumes can be connected to several EC2 instances and on-premises resources.

How Does EFS Work?

- EFS can be created using the EC2-Instance where it will be created in a specific region and distributed across multiple availability zones for the purpose of high availability and durability. You can choose the EFS based on the I/Ops you are going to perform.

- Once the EFS is created you need to set up mount targets which will provide the connectivity to your EFS file system. An EFS (Elastic File System) Mount Target is a network endpoint that enables an EC2 instance to communicate with an EFS file system within a Virtual Private Cloud (VPC). Following are some of the resources which you can mount on EFS.

- Amazon EC2.
- Amazon ECS.
- Amazon EKS.
- AWS Fargate.
- AWS lambda and some other servers.



Different Storage Classes in AWS EFS

Standard storage class

- This is the default storage class for EFS.
- The user is only charged for the amount of storage used.
- This is recommended for storing frequently accessed files.

Infrequently Accessed storage class(One Zone)

- Cheaper storage space.
- Recommended for rarely accessed files.
- Increased latency when reading or writing files.
- The user is charged not only for the storage of files but also charged for read and write operations.

Use Cases Of EFS

- **Secured file sharing**: You can share your files in every secured manner and in a faster and easier way and also ensures consistency across the system.

- **Web Hosting**: Well suited for web servers where multiple web servers can access the file system and can store the data EFS also scales whenever the data incoming is increased.

- **Modernize application development**: You can share the data from the AWS resources like ECS, EKS, and any serverless web applications in an efficient manner and without more management required.

- **Machine Learning and AI Workloads**: EFS is well suited for large data AI applications where multiple instances and containers will access the same data improving collaboration and reducing data duplication

When To Choose Amazon EFS?

Amazon EFS is suitable for the following scenarios:

- **Shared File Storage:** If the multiple EC2-Instances have to access the same data. EFS management of shared data and ensures consistency across instances.
- **Scalability:** EFS can increase and decrease its storage capacity depending on the incoming data. If you don't have an idea how much data is going to come to the store then you can use the Amazon EFS.
- **Simplified Data Sharing:** If different applications want the same data to use in a collaborative manner then you can choose the Amazon EFS. EFS can share large datasets across a group of instances.
- **Use with Serverless Applications:** Amazon EFS is well suited for the service like serverless computing services like some of the examples AWS lambda, EFS, and so on.
- **Pay-as-You-go-Model:** If your application is having unpredictable storage growth then there is no need of paying upfront or no need of any prior commitments. You pay only for the storage that you are going to use.